

The Hale Desert Martin House

LIST OF MATERIALS

WOOD

1/4-in x 2-ft Bcx Pine Sanded Plywood	\$ 6.55	3	\$ 19.65
1-in x 12-in x 8-ft White Pine Board	\$ 16.89	1	\$ 16.89
5/8-in x 5-1/2-in x 6-ft Cedar Fence Picket	\$ 2.98	2	\$ 5.96

SAND

50-lb All-purpose Sand	\$ 6.48	1	\$ 6.48
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INSULATION

R- 21 Reflective Insulation (16-in W)	\$ 19.97	1	\$ 19.97
general purpose insulation	\$ 7.48	1	\$ 7.48

LINER

Extra Small Moving Box (10x12x15)	\$ 1.28	1	\$ 6.40
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HARDWARE

Safety Gate Hook	\$ 2.96	2	\$ 5.92
Hinges	\$ 4.66	1	\$ 4.66

Subttl	\$ 93.41
Sales Tax	\$ 7.74
Total	\$ 101.15

DIMENSIONS

OUTER BOX

12 5/8" wide (front and back) x 14" long (sides). 13 1/4" tall (13.5" with roof lid)

Roof Lid – hinged. Plywood and cedar slats – 14 1/2" wide by 15" long front to back. 2 1/2" overhang soffit.

Back and sides standard pine board. Front 1/4" plywood.

Option – front pine board instead of plywood?

Option – Cedar can be used instead of pine, better quality wood, lighter, but much more expensive.

INNER BOX (NEST CHAMBER)

Interior 5 1/2" wide x 6.2" long. 7 1/4" depth from entrance shelf. 10 1/4" tall.

Entrance shelf 2 1/2" 'tunnel bridge' from entrance to nest chamber.

5 1/2" cedar fence slats used for width. Plywood sides and hinged ceiling.

Plywood sides are length of outer box for stability of nest chamber to center it in the box.

*** In lieu of sketch, see finished work photos on following pages. I built it in my shop.

INSULATION DETAILS

The interior of the Outer Box is lined with reflective wall insulation – sides, top, bottom.

The Inner Box (Nest Chamber) is placed in a 10"x12" half-cut cardboard moving box

The interstitial between the next box and cardboard box is filled with sand about 2/3rd to the top.

General insulation is packed in over the sand on both sides and the back wall.

A gap in the roof soffit on each side allows for airflow and heat transfer out of the box.

INSTALLATION METHOD

The weight of the entire apparatus with sand is 36.5 lbs. That's heavy. The box will need a strong installation structure, on the back of the box, perhaps 2x4s or 2x6 with angle supports to the bottom. Maybe a steel support? Depends on what you are attaching it to, what kind of pole. I envisioned these attached to utility poles out in the desert. They do that in NM, nesting platforms on utility poles, for ravens so they won't nest at the top of the pole in the wires. Or, it could attach on the bottom of the box using a cross-beam platform set on top of a pole.

REASONING BEHIND THE DESIGN

The Hale desert martin house is a box inside a box inside a box. A nest chamber box placed in a cardboard box filled with sand, housed inside a wooden box lined with thin wall insulation. Have you ever been to the beach (or desert) and stuck your toes in the hot sand? The sand is nice and cool right below the surface! Sand has amazing insulation properties. That should cool down the nest chamber. The side wall thin bubble reflective insulation further protects the interior from overheating. I kept the outside unpainted, natural wood, thinking that would be more attractive to martins that are not accustomed to manmade nest sites. The front of the box has lengthwise strips of wood to simulate the saguaro cactus ribbing. The tops of both outer and inner boxes are hinged for easy access to the nest chamber. The front of the outer box is hinged for easy access to remove/replace the cardboard box containing the nest chamber box and sand.

I'm not an engineer or expert in thermodynamics. I don't like plywood for birdhouses, but it solved several design challenges. A coating of water repellant would probably be wise, you don't want the cardboard box or sand getting wet. My hope, if nothing else, is that this design might be incorporated directly or combined with others for the benefit of desert Purple Martins (that I have yet to actually observe). I love my martins here in Texas, they are awesome birds, we can't wait for them to arrive at our house every year. They bring joy and excitement like no other bird. Progne subis is truly a wonder.

Overall View, front and side.



Outer Accesses – front panel, roof lid. Note insulated lining. Nest chamber sitting in half moving box and be pulled out for end of season cleaning, replacement.



Inside box nest chamber view, closed and with ceiling raised.



Sand box pulled forward.



Cutaway view of nest chamber entry bridge.



DESIGN UPDATE

03/05/2023

FRONT FASTENERS	Replace Phillips head screws fasteners with eye-screws for easy twist open/close of the outer box front panel.
ROOF SOFFIT VENT	Add 5/16" vent holes on front top corner or "dog-ear" notch the top corners to allow for air flow under the roof for better heat transfer.
ROOF ADD A PERCH	Add a stick perch extension one side of roof extending 2-3 inches over the front. Appearance as a natural stick or bud jutting out on top. Martins seem to like perching near the nest entrance.
ROOF GUARD LINE	Add a string or small chain line to keep roof from falling back too far when opening.
ROOF HINGE SCREWS	Replace the small 1/2" hinge screws with #6 3/4" screws for more secure attachment.
NEST CHAMBER BOX	CEILING - Extend ceiling panel 1/2" to cover completely the gap space from the cardboard box and insulation to the outer wall. BRIDGE - Extend entry bridge 1/2" to cover completely the gap space from the cardboard box and insulation to the outer wall.
NEST CHAMBER SIDES	-option- use cedar panels instead of plywood. 1/4" tongue and groove cedar panels.
OUTER INSULATION (?)	-option- add layer of insulation on south and west outer sides with plywood or cedar covering?
WASP DEFENSE	Rub soap on the ceilings of the outer box lid and nest chamber panel to discourage wasps.

- Updated overall views. Additions: roof perch, vent holes, eye screws



- Front fasteners replaced with eye screws for easy twist on/off. Soffit vent holes added (top front corners). Aesthetically, I think I would prefer a 45-degree angle dog-ear cut instead. Looks like insect holes here.



- Roof guard line. Don't want roof falling back too far or snapping off the hinges. Notice nest chamber ceiling extends all the way to the outer box inside wall.



- Nest Chamber Box – ceiling and entry bridge extended $\frac{1}{2}$ " to cover the gap from the cardboard box and insulation to the inside wall of the outer box. I replaced the cedar bridge piece with a plywood piece instead, matching the exterior appearance, just seemed a better fit looking in and gives a slight "step" inside from the entrance hole.

